

Cross-Entropy

math-foundations 37-cross-entropy

1 Definitions

Definition 1 (Discrete cross-entropy). Let p, q be probability mass functions on a countable alphabet $\mathcal{X}$, with $q(x) > 0$ whenever $p(x) > 0$. The *cross-entropy* of p with respect to q is

$$H(p, q) = - \sum_{x \in \mathcal{X}} p(x) \log q(x) = -\mathbb{E}_{X \sim p}[\log q(X)].$$

We adopt $0 \log 0 = 0$. If there exists $x \in \mathcal{X}$ with $p(x) > 0$ and $q(x) = 0$, then $H(p, q) = +\infty$.

Definition 2 (Continuous cross-entropy). For probability densities f_p, f_q on $\mathbb{R}^d$ with $f_q > 0$ on the support of f_p ,

$$H(p, q) = - \int_{\mathbb{R}^d} f_p(x) \log f_q(x) dx.$$

Definition 3 (Kullback–Leibler divergence). With the same hypotheses,

$$D_{\text{KL}}(p \parallel q) = \sum_x p(x) \log \frac{p(x)}{q(x)} = H(p, q) - H(p),$$

where $H(p) = -\sum_x p(x) \log p(x)$ is Shannon’s entropy.

2 Gibbs’ inequality

Lemma 4 (Logarithm bound). *For all $z > 0$,*

$$\ln z \leq z - 1,$$

with equality if and only if $z = 1$.

Proof. Let $f(z) = z - 1 - \ln z$ on $(0, \infty)$. Then $f'(z) = 1 - 1/z$, so $f'(z) < 0$ for $z \in (0, 1)$ and $f'(z) > 0$ for $z > 1$, with $f'(1) = 0$. Hence f has a strict global minimum at $z = 1$ with $f(1) = 0$. Therefore $f(z) \geq 0$ everywhere, with equality only at $z = 1$. $\square$

Theorem 5 (Gibbs’ inequality). *For any two distributions p, q on a common alphabet $\mathcal{X}$ with q absolutely continuous with respect to p on the support of p ,*

$$H(p, q) \geq H(p),$$

with equality if and only if $p(x) = q(x)$ for all x in the support of p .

Proof. We show the equivalent statement $D_{\text{KL}}(p \parallel q) \geq 0$. Restrict the sum to $\mathcal{X}_0 = \{x : p(x) > 0\}$; on this set $q(x) > 0$ by hypothesis, so $q(x)/p(x)$ is a positive real number.

Apply Lemma 4 with $z = q(x)/p(x)$ for each $x \in \mathcal{X}_0$:

$$\ln \frac{q(x)}{p(x)} \leq \frac{q(x)}{p(x)} - 1.$$

Multiplying both sides by $p(x) \geq 0$ preserves the inequality, and summing over $x \in \mathcal{X}_0$ gives

$$\sum_{x \in \mathcal{X}_0} p(x) \ln \frac{q(x)}{p(x)} \leq \sum_{x \in \mathcal{X}_0} (q(x) - p(x)) = \left(\sum_{x \in \mathcal{X}_0} q(x) \right) - 1 \leq 1 - 1 = 0,$$

where the last inequality uses $\sum_{x \in \mathcal{X}_0} q(x) \leq \sum_{x \in \mathcal{X}} q(x) = 1$. Converting to an arbitrary log base by dividing through by $\ln(\text{base})$ does not change the sign, so

$$-D_{\text{KL}}(p \parallel q) = \sum_{x \in \mathcal{X}_0} p(x) \log \frac{q(x)}{p(x)} \leq 0,$$

i.e. $D_{\text{KL}}(p \parallel q) \geq 0$. By the decomposition $H(p, q) = H(p) + D_{\text{KL}}(p \parallel q)$, this gives $H(p, q) \geq H(p)$.

Equality case. Suppose $H(p, q) = H(p)$, so $D_{\text{KL}}(p \parallel q) = 0$. Each summand $p(x) \ln(q(x)/p(x))$ is bounded above by $q(x) - p(x)$ (Lemma 4), and we must have equality in both the Lemma-4 step and the marginal step. Equality in Lemma 4 forces $q(x)/p(x) = 1$ for every $x \in \mathcal{X}_0$, i.e. $q(x) = p(x)$ on the support of p . The marginal step requires $\sum_{x \in \mathcal{X}_0} q(x) = 1$, which is now automatic since $\sum_{x \in \mathcal{X}_0} q(x) = \sum_{x \in \mathcal{X}_0} p(x) = 1$. Conversely, if $p = q$ on $\mathcal{X}_0$ then every term in D_{KL} vanishes, so $H(p, q) = H(p)$. $\square$

Corollary 6 (Decomposition). *For any p, q as above,*

$$H(p, q) = H(p) + D_{\text{KL}}(p \parallel q),$$

and minimizing $H(p, q)$ over q (with p fixed) is equivalent to minimizing $D_{\text{KL}}(p \parallel q)$ over q .

Proof. Direct from the definitions: $-\sum_x p(x) \log q(x) = -\sum_x p(x) \log p(x) + \sum_x p(x) \log \frac{p(x)}{q(x)}$. The second statement follows because $H(p)$ does not depend on q . $\square$

Example 7 (Fair coin, biased code). Let $p = (\frac{1}{2}, \frac{1}{2})$ and $q = (0.9, 0.1)$ on $\mathcal{X} = \{0, 1\}$. Then $H(p) = 1$ bit, and

$$H(p, q) = -\frac{1}{2} \log_2 0.9 - \frac{1}{2} \log_2 0.1 \approx 1.737 \text{ bits},$$

so $D_{\text{KL}}(p \parallel q) \approx 0.737$ bits — the per-symbol coding overhead of using a code matched to q on data drawn from p .